

ANIRUDH IYENGAR

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Education

University of California, Berkeley

Expected Graduation: May 2028

BS in Electrical Engineering and Computer Science (EECS) | Physics Minor

- Relevant Courses: Signals & Systems, Circuits & Devices; Structure & Interpretation of Computer Programs; Linear Algebra & Differential Equations; Data Structures & Algorithms; Mechanics; E&M; Microelectronics Devices

Experience

NeuroTechX

Sept 2026 - Present

Incoming Hardware Consultant

San Francisco, Ca

- Analog front-end design and hardware integration for a high-density EEG device built on top of the Open Ephys acquisition platform

LOCI Lab @ CMU (PI: Dr. Andrew Spielberg)

May 2026 – Present

Robotics Intern

Pittsburgh, PA

- Designed and validated a novel, camera-free ultrasonic tactile-sensing platform for real-time robotic manipulation; developed mixed-signal PCBs in Altium and C++ firmware for synchronized 1 MHz, 5-cycle excitation and 12-bit waveform acquisition at 20 MS/s, enabling full transmit-and-echo capture for contact and material sensing
- Built multiphysics finite-element models in ANSYS Mechanical and Cosmos to simulate piezoelectric deformation, stress distribution, resonant behavior, and ultrasonic propagation; performed parametric sweeps across sensor geometry, material properties, excitation frequency, and applied contact force to optimize sensor sensitivity and mechanical design

Sensorimotor Neural Engineering Lab (PI: Dr. Preeya Khanna)

Feb 2026 – Present

Undergraduate Researcher

Berkeley, CA

- Developing a low-cost, multi-sensor wireless wearable for out-of-clinic hand tracking in stroke rehabilitation: fabricating liquid-metal e-skin bend sensors (mean absolute error $\sim 2^\circ$ benchtop, $\sim 5^\circ$ on-hand) and integrating IMUs for wrist kinematics; designing analog signal conditioning circuits and a Raspberry Pi data acquisition pipeline with Bluetooth streaming
- Analyzing fully sensorized glove kinematic data from healthy and stroke-survivor populations (in collaboration with UW) to identify principal patterns of hand motion via PCA and dimensionality reduction; determining the minimum sensor subset needed to capture dominant hand kinematics, directly informing optimal sensor placement for the wearable system

University of Louisville Sport Science

May 2023 – June 2025

Machine Learning Researcher

Louisville, KY

- Developed a YOLOv8-based computer vision system to extract biomechanical kinematics of baseball athletes; validated model performance against ground-truth data, improving injury detection accuracy by 12% over manual baselines
- Built end-to-end Python + OpenCV + NumPy pipelines for video preprocessing, tracking, and feature extraction; automated data collection and analysis workflows, improving keypoint extraction throughput by 20%

Projects and Campus Involvement

NeuroTech @ Berkeley

Aug 2025 - Present

- Designed, simulated, and validated a multi-stage EMG biosignal acquisition system: implemented analog band-pass filtering (20–500 Hz), mid-supply biasing, and Schottky-clamped ADC protection; verified gain, noise performance, and signal integrity via oscilloscope and FFT analysis in LTSpice
- Developing an Arduino Mega-based amplification shield for a 64-channel ultrasonic phased-array; designing MOSFET driver circuits to amplify 40 kHz logic signals to 17 Vpp for haptic/levitation applications; PCB layout in Altium with modular 16-way IDC connectors and 250 Kbaud UART for daisy-chained signal distribution

LiDAR Landing System

Sept 2025 - Dec 2026

- Awarded a \$1,500 NASA research grant to investigate LiDAR-based autonomous rocket landing systems
- Designed and fabricated a custom PCB in KiCAD integrating flight sensors (IMU, barometer, rangefinder) for terrain data collection; validated board functionality with multimeter and oscilloscope during bring-up
- Collected experimental data to train a custom OpenCV computer vision model for lunar terrain recognition and autonomous navigation

Space Enterprise @ Berkeley

June 2026 - Present

- Transitioning the avionics bay and harness configuration to a distributed Ground Support Equipment (GSE) system, developing its networking infrastructure, and repairing heavily used GSE boards.

Technical Skills

Hardware & EE Tools: Altium Designer, KiCAD, LTSpice, PCB/Circuit Design (Analog, Digital, Power), Oscilloscopes, Signal Generators, Multimeters, STM32 Microcontrollers, Arduino, Raspberry Pi, Piezoelectric Transducers, Motor Drivers

Software & Languages: Python, C++, C, MATLAB, Verilog, Java, JavaScript, SQL, HTML5, CSS

Design & Simulation: CFD, CAD/CAM, Ansys, SolidWorks, FEA, AutoCAD, Revit, Microsoft Office